Biochemistry Cume
March 1, 2002

One of the most exciting scientific achievements of all time was the recent sequencing of the human genome. Many other genomes have been sequenced and more are being sequenced all the time.

1. One genome that is currently being sequenced is that of the pufferfish. Researchers who study the human genome are very interested in this genome because it is quite compact with little "junk" DNA. Explain how this genome can be used to interpret the human genome. 30 points

2. The basic method for sequencing genomes is often called "shotgun sequencing." Briefly explain the basics of how one starts with an intact genome of DNA and ends up with an assembled sequence in the computer. Do not explain how the actual DNA sequencing itself works (see next question). 30 points

3. The common method used for DNA sequencing itself is so-called Sanger method, after Fred Sanger who developed it. Explain briefly how it works. 30 points

4. For many years the standard method for detecting DNA fragments was the autoradiogram, a piece of x-ray film next to a gel with radioactively labeled DNA. While still a useful technique in some ways, genome centers need high-throughput for sequencing and autoradiograms are slow. How is the DNA sequencing ladder labeled and detected in the sequencing machines that they use? 10 points